

Task

Analyze a dataset using Numpy, Pandas, Matplotlib, and Seaborn. Display the data, provide a brief description, plot histograms for two features, and create a pairwise scatter plot to identify relationships.

Load the dataset

Subtask:

Load a suitable dataset using pandas.

Import the pandas library and load a suitable dataset from a publicly available URL into a pandas DataFrame.

```
import pandas as pd

# Load a sample dataset (Iris dataset from UCI Machine Learning Repository)

url =
"https://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data"

col_names = ['sepal_length', 'sepal_width', 'petal_length',
'petal_width', 'class']

df = pd.read_csv(url, header=None, names=col_names)

# Display the first few rows of the DataFrame

display(df.head())
```

	sepal_length	sepal_width	petal_length	petal_width	class
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

Display and describe the data

Subtask:

Display the first few rows and provide a brief description of the data using pandas and numpy.

```
display(df.head())
```

```
display(df.info())
```

```
display(df.describe())
```

	sepal_length	sepal_width	petal_length	petal_width	class
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   sepal_length    150 non-null   float64
1   sepal_width     150 non-null   float64
2   petal_length    150 non-null   float64
3   petal_width     150 non-null   float64
4   class           150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000